

END SEMESTER ASSESSMENT (ESA) - May 2023**UE21CS242A - Web Technologies****Total Marks : 100.0**

1.a. What is HTTP? Explain the structure of HTTP request message. (4.0 Marks)

- 1.b. Create an HTML form with the following controls:
1. A text box to collect the user's name
 2. A multiline input field to collect user's address
 3. Three checkboxes, one each for the following items:
 1. wireless mouse
 2. Mobile Charger
 3. USB
 4. A collection of three radio buttons for Payment option that are labelled as follows:
 1. Net Banking
 2. Google Pay
 3. Cash on Delivery
 5. Submit and reset Button

(7.0 Marks)

1.c. What are the 3 ways of including CSS on an HTML page? Give Example (6.0 Marks)

1.d. Explain the concept of hoisting in JavaScript? How can we avoid hoisting in JavaScript? (3.0 Marks)

2.a. Create a HTML form with 2 input fields and 2 buttons invoke click event on the buttons to calculate multiplication and division of two numbers by taking form input from user.

(6.0 Marks)

1st Number :

2nd Number:

2.b. Write jQuery code to perform the following:

1. For the last paragraph within the div with id "colortext", set the color to yellow.
2. On moving the mouse over an h1 element, the text font size should be increased by 3 times
3. When the first li element with class name "liclass" on the page is clicked, it fades out in 2 seconds and then fades in 3 seconds

(6.0 Marks)

2.c. Explain event.PreventDefault, event.stopPropagation with example (4.0 Marks)

2.d. What are different ways through which we can access the DOM elements? Give any four with example (4.0 Marks)

3.a. With a neat diagram explain component life cycle. (6.0 Marks)

3.b. Given an array of names of 3 fruits of your choice, render an unordered list with each fruit name as a list item. (4.0 Marks)

3.c. What is the difference between DOM Events and Synthetic Events. Give example? (4.0 Marks)

3.d. List out the differences between stateful and stateless components (6.0 Marks)

4.a. What are buffers in Node.js? With help of syntax explain any 2 buffer operations (5.0 Marks)

4.b. Explain File system Module in NodeJS with an example (6.0 Marks)

4.c. Imagine you have a MongoDB database "myDB" which has collection "students" with fields student_ID, student_Name, Semester, Department and Contact_no. Write a program to connect to MongoDB database in Node.js to find data of all students who study in CSE department.
Note: Only write Server Side code (5.0 Marks)

4.d. Create a module that calculates the area of a circle and circumference of a circle.
Give the radius, export the module and print the area of circle and circumference of a circle on the console. (4.0 Marks)

5.a. What is RESTful API? Explain any 4 design specification/constraints of REST API. (4.0 Marks)

5.b. Explain the role of express middleware function. What are the different types of middleware an express application can use? (6.0 Marks)

5.c. Write a program to upload a file under the upload folder in the same directory as the JS file with proper error handling.
Note: Write only the server JS code. HTML/client code is not required. (6.0 Marks)

5.d. Explain Error handling in ExpressJS

(4.0 Marks)